COMP0199

1 Sequences of Functions

$(f_n)_{n=0}^{\infty}$ or (f_n) denotes the sequence of functions where $f_n : A \to \mathbb{R}$ for all $n \in \mathbb{N}$.

Sequence of Functions An assignment of a function $f_n : A \to \mathbb{R}$ to each $n \in \mathbb{N}$.

Hereafter, assume A as the domain / target interval.

1.1 Background

For numbers, the sequence u_n tends to l iff.

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : n > N \Rightarrow |u_n - l| < \varepsilon$$

A sequence of numbers u_n is Cauchy if

$$\forall \varepsilon > 0, \exists N > \mathbb{N}, \forall m, n \geq N : |u_m - u_n| < \varepsilon$$

A sequence is Cauchy iff. it is convergent (in *complete* metric spaces). Note that the Cauchy Criterion does not refer to the limit itself.

1.2 Pointwise Convergence

If $\lim_{n\to\infty}(f_n(a))$ exists and is finite for all $a \in A$,
we can define the *limit function* $f : x \mapsto \lim_{n\to\infty}(f_n(x))$.
$(f_n(a))_{n\in\mathbb{N}}$ is said to **converge *pointwise*** towards f.

Pointwise Convergence The sequence $(f_n)_{n\in\mathbb{N}}$ converges pointwise to $f : A \to \mathbb{R}$ if

$$\forall x \in A, \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n > N : |f(x) - f_n(x)| < \varepsilon$$

I.e., $\forall x \in A, f_n(x) \to f(x)$ as $n \to \infty$.

A sequence of functions (f_n) is pointwise Cauchy if $(f_n(x))$ is Cauchy for all x.
A sequence is pointwise Cauchy iff. it converges pointwise.

1.3 Uniform Convergence

Uniform Convergence The sequence $(f_n)_{n\in\mathbb{N}}$ **converges *uniformly*** to f on the interval A if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall x \in A, \forall n > N : |f(x) - f_n(x)| < \varepsilon$$

I.e., $\sup_{x\in A}|f_n(x) - f(x)| \to 0$ as $n \to \infty$.

Uniform convergence implies pointwise convergence, to the same limit. (It is *stronger*.)

Uniformity is preserved:
The limit f continuous if all f_n is continuous and (f_n) converges uniformly.

Integrals are preserved:
If (f_n) converges uniformly to f on $[a, b]$, then

$$\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b f(x)dx$$

Derivatives are more complicated:
If $(f_n(x_0))$ converges for some $x_0 \in [a, b]$ and $(\boldsymbol{f_n'})$ converges uniformly on $[a, b]$, then

$$\lim_{n\to\infty} f_n'(x) = f'(x) \qquad (\text{for } x \in [a, b])$$

...and the convergence of (f_n) is also uniform.

Note: the uniform convergence on the sequence of derivatives converges the "shape" of the function, while the single-point convergence on the sequence itself asserts that the solution is uniquely well-defined. (Consider $f_n(x) = n + x$.)

2 Series of Functions

$\sum f_n$ denotes the series of functions, i.e., the sum of the sequence (f_n).

2.1 easy stuff

$f = \sum f_n$ is more specifically $f : x \mapsto \sum_{n=\infty}^{\infty} f_n(x)$.
Clearly, this (infinite) sum may be undefined.

The partial sums of a series is a sequence itself: $\left(\sum_{k=0}^{n} f_k(x)\right)_n$

$\sum f_n$ **converges *pointwise*** on A if the sequence of *partial sums* converges pointwise on A.

Weierstrass M-Test A test for determining whether a series of functions converges *normally*:
$\sum f_n$ **converges *normally*** on A
if $\forall n \in \mathbb{N}, \forall x \in A : |f_n(x)| \leq M_n$
for $M_n \geq 0$, $\sum M_n$ converges.

Note that *normal convergence* implies ***uniform*** and ***absolute*** **convergence**.
The Weierstrass M-test may be seen as the Cauchy Criterion for uniform convergence.
Furthermore, the x-dependence is eliminated by effectively taking the suprema:

$$\sup_x \left| \sum_{n=N}^{\infty} f_n(x) \right| \leq \sum_{n=N}^{\infty} \sup_x |f_n(x)| = \sum_{n=N}^{\infty} M_n \to 0$$

- Does not help find the actual limit.
- Main tool used to test for uniform convergence.

2.2 Power Series

Power Series Series of functions of the form

$$\sum_{n \geq 0} a_n (x-b)^n$$

- "centered" around b
- effectively an infinite polynomial

A power series always converges for certain x:
- trivially true for $x = b$,
- then there exists a convergent neighborhood around b.

Radius of Convergence Some $r \geq 0$ where a power series converges uniformly on any closed interval $A \subset (b-r, b+r)$.
Divergence when $x \notin [b-r, b+r]$; unknown when $x = b \pm r$.
Can be infinite for series convergent everywhere.

D'Alembert's Ratio Test A test to find the radius of convergence r of a power series.
For some $\sum_{n \geq 0} a_n (x-b)^n$, consider the limit of the sequence of a-ratios $L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$:

$$r = \begin{cases} \frac{1}{L} & \text{if } L \neq 0 \\ \infty & \text{if } L = 0 \\ 0 & \text{if } L = \infty \end{cases}$$

I.e., convergence when $|x - b| < \frac{1}{L}$.

2.3 Taylor Series

Taylor Series are a special case of power series.
They converge (within the radius) to arbitrary functions.

For a function f infinitely differentiable at $b \in \mathbb{R}$, its Taylor Series centered at b is

$$\sum_{n \geq 0} \frac{f^{(n)}(b)}{n!} (x-b)^n$$

- converges to f in $(b-r, b+r)$
- Maclaurin Series are a special case where $b = 0$

2.3.1 Well-known Functions

$$\begin{aligned}
e^x &= \sum_{n \geq 0} \frac{x^n}{n!} &&= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\
\cos(x) &= \sum_{n \geq 0} \frac{(-1)^n}{2n!} x^{2n} &&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \\
\sin(x) &= \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)!} x^{2n+1} &&= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \\
\ln(1+x) &= \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} x^n &&= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots && \text{for } |x| < 1, x = 1 \\
\frac{1}{1-x} &= \sum_{n \geq 0} x^n &&= 1 + x + x^2 + x^3 + \dots && \text{for } |x| < 1 \\
\arctan(x) &= \sum_{n \geq 0} \frac{(-1)^n}{2n+1} x^{2n+1} &&= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots && \text{for } |x| < 1
\end{aligned}$$

2.3.2 Approximations

The degree-n Taylor polynomial of f centered at b is just the series truncated after $n+1$ terms:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(b)}{k!} (x-b)^k$$

$f(x) \approx T_n(x)$ for $x \approx a$.

$R_n(x) = f(x) - T_n(x)$ is the tail of the terms in the series.
There are numerous ways to express this sub-series beyond a simple summation.

Lagrange Form of the Taylor Remainder

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-b)^{n+1}$$

where ξ is a function of x and is strictly between b and x (open interval).

The derivation of such a term that "captures" an entire series comes from the Mean Value Theorem; the heavy lifting is done by ξ, which is a function itself and is *not* computable, but is bounded and thus provides an upper bound for $R_n(x)$. Let $M = \max_{\xi \in (b,x)} \left| f^{(n+1)}(\xi) \right|$:

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-b|^{n+1}$$

Since M is constant, we have, for $x \to b$:
- $R_n(x) \approx |x-b|^{n+1}$, or more specifically,
- $R_n(x) = O((x-b)^{n+1})$
- $R_n(x) = o((x-b)^n)$

3 Linear Algebra

Assume U, V, and W to be vector spaces.
Assume vector spaces to be over the field of $\mathbb{R}$.
Assume $+$ over functions be *pointwise addition* – $(f+g)(x) = f(x) + g(x)$.

3.1 easy stuff

Linear Independence A subset $S = \{v_1, ..., v_k\} \subseteq V$ is linearly independent if
$\sum_{i=0}^{k} a_i v_i = \mathbf{0} \implies$ all $a_i = 0$

Alternatively, the map $(a_1, ..., a_k) \mapsto a_1 v_1 + ... + a_k v_k$ is injective.

Linear A map $f : V \to W$ is linear iff.
additivity $f(u+v) = f(u) + f(v)$ and
homogeneity $f(\lambda v) = \lambda f(v)$
are preserved (for arbitrary $u, v \in V$ and $\lambda \in \mathbb{F}$).

Additivity and homogeneity are commonly combined into $f(\lambda u + v) = \lambda f(u) + f(v)$.

Linear Map A *homomorphism* (see below) of a *vector space*.

Basis (…of a vector space V) is a subset $B \subseteq V$ that is linearly independent and *spans* V.

Direct Sum A structure-*propagating* Cartesian product[2] with operations defined element-wise. Notated with "oplus" $\oplus$.

The above is more specifically an **external** direct sum; an **internal** direct sum *conditionally* exists for *sub*structures (of a common abelian-group-like structure) iff. their intersection is *trivial*;
For a structure S with substructures A and B,
$A \oplus B$ is well-defined[1] ("A and B are in direct sum") iff. $A \cap B = \{\mathbf{0}\}$.

Decomposition Statements The expression $S = A \oplus B$ asserts that A and B are in direct sum and that their sum is exactly S.

$$S = A \oplus B \iff \text{“}A \oplus B\text{” } \textbf{and } S = A + B$$

Note: with internal direct sums, the $A \oplus B$ notation has overloaded meanings: it denotes the same sum set as $A + B$ while simultaneously *asserting* that $A \cap B = \{\mathbf{0}\}$.

Alternative definitions of internal direct sum:
Substructures A and B of a common S are in direct sum iff. every element of $T = A + B$ can be written in exactly one way as the sum of some $a \in A$ and $b \in B$. I.e.,

$$A \oplus B \Longleftrightarrow \forall t \in T\ \exists!(a, b) \in A \times B : t = a + b$$

Again, $S = A \oplus B$ if also $T = S$. A more general version, assuming $V = \sum_i V_i$:

$$V = \bigoplus_i V_i \iff \forall v \in V, \exists!(v_i)_i \in \bigoplus_i V_i : v = \sum_i v_i$$

Note that the second $\oplus$ denotes the external direct sum.
The generalized intersection criteria: $\bigoplus_i V_i \iff \forall i : V_i \cap \sum_{j \neq i} V_j = \{\mathbf{0}\}$.

Note: the *direct product* is a structure-propagating Cartesian product;
the external direct sum is a direct product with the restriction that the resulting tuples can only have finitely many non-zero elements (which matters if there are infinite operands).

If $(u_1, ..., u_n)$ is a basis of U and $(v_1, ..., v_m)$ is a basis of V, then $U \oplus V$ iff. $(u_1, ..., u_n, v_1, ..., v_m)$ are linearly independent. (U and V are in the same ambient space.)
The internal direct sum is associative; the external direct sum is associative up to isomorphism.

Composition For functions $f : B \to C$ and $g : A \to B$,
composition is a binary operator where $f \circ g : A \to C$, $x \mapsto f(g(x))$.

Composition verifies (for arbitrary functions f, g, h):
- associativity $f \circ (g \circ h) = (f \circ g) \circ h$,
- identity $f \circ \mathrm{id} = \mathrm{id} \circ f = f$, and
- distributivity (over $+$)
  right $(f + g) \circ h = (f \circ h) + (g \circ h)$ and
  left $f \circ (g + h) = (f \circ g) + (f \circ h)$ **only if** f is *linear*.

$(\mathrm{End}(V), +)$ is an abelian group;
$(\mathrm{End}(V), +, \circ)$ is a ring.
$\mathrm{End}(V) := \{f : V \to V \mid f \text{ is linear}\}$; see below section on Endomorphisms.

Note that, for linear f (so symmetric distributivity), we obtain the properties as matrix multiplication.

Dimension The dimension of a vector space V, notated $\dim V$, is the cardinality of (any) basis of V.
Note that the Steinitz exchange lemma guarantees any two bases of V have the same cardinality.
$\dim \{0\} = 0$.

For a function $f : A \to B$:
Injectivity $\forall x, y \in A : f(x) = f(y) \Rightarrow x = y$ i.e., $\forall y \in \mathrm{im}\, f, \exists! x \in A : f(x) = y$
Surjectivity $\forall y \in B, \exists x \in A : f(x) = y$ i.e., $\mathrm{im}\, f = B$
Bijectivity Injectivity *and* Surjectivity.

3.2 Linear Maps

Assume f, g, h to be linear maps, and by default $f : V \to W$.

Image $\mathrm{im}\, f := \{f(v) \mid v \in V\} \quad = f(V) \subseteq W$
Kernel $\ker f := \{v \in V \mid f(v) = \mathbf{0}\}$
Rank $\mathrm{rank}\, f = \dim \mathrm{im}\, f$
Rank-Nullity Theorem $\mathrm{rank}\, f + \dim \ker f = \dim V$

Note that Rank is only defined for images of finite dimension, and
Rank-Nullity Theorem assumes V to have finite dimension too.

Kernel criterion for Injectivity f is injective iff. $\ker f = \{\mathbf{0}\}$.

Clearly, injectivity implies $\dim W \geq \dim V$. (Can be proved using the Rank-Nullity Theorem.)

Matrix Column Rank criterion for Injectivity f is injective iff. for a matrix M representing f, its columns are linearly independent.

Clearly, surjectivity implies $\dim V \geq \dim W$. (Can be proved using the Rank-Nullity Theorem.)

Note that any linear map can be made surjective trivially by setting its codomain (W) to its image.

Isomorphism Both-way-*structure-preserving* bijections.

Examples of structure properties to preserve: linearity, group operation.
The *structure* of vector spaces is linearity, and linear maps are, well, linear bidirectionally, so all **bijective linear maps are isomorphisms**.

Note[2]: The concept of *structure-preserving maps* come from category theory, where categories explicitly define its such maps, called *morphisms*; vector spaces' morphisms are linear maps by definition, so bijective linear maps are isomorphisms :shrug:.

Clearly, bijectivity implies $\dim V = \dim W$.
Intuitive way to express bijectivity: $\forall w \in W, \exists! v \in V : f(v) = w$.

Two structures are *isomorphic* iff. there exists an isomorphism between them.
For V and W over the same field, $\dim V = \dim W$ iff. they are isomorphic.

Matrix representations of a bijection must be square.
A matrix is invertible iff. it represents a bijection.

3.3 Endomorphisms

Endomorphism A map $f : V \to V$; same domain and codomain.
Automorphism An endomorphism that is also an isomorphism.

For endomorphisms, **injectivity, surjectivity, and bijectivity are equivalent** (in finite dimension).

f is bijective iff. it maps a basis to a basis.

3.3.1 Change of Basis

Consider an endomorphism f of V represented by matrix M in the basis B_1.

$N = P^{-1} M P$ is the matrix representing f in B_2 where P is a matrix whose columns are the coordinate vectors of B_2 in B_1.

Transition Matrix (aka. change-of-basis matrix) what P is above.

Observe that if $P^{-1} M P = N$, then $P N P^{-1} = M$.

4 Matrix Reduction

Assume V denotes some finite-dimensional vector space over $\mathbb{F}$ ($= \mathbb{R}$ or $\mathbb{C}$, as specified).
Assume $T : V \to V$ denotes some endomorphism of V.
Assume $M \in \mathcal{M}_{n,n}$ is the matrix representation of any such T.

Observe that as T is an endomorphism and $n = \dim V$, we can always represent it as some square matrix.

4.1 Abstract ahh Stuff

Invariant Subspace (aka. stable subspace) $U \subseteq V$ is an invariant subspace under T if $T(U) \subseteq U$.
I.e., $\forall u \in U : T(u) \in U$.

Observe that V and $\{\mathbf{0}\}$ are always invariant subspaces under *any* T;
$\ker T$ and $\operatorname{im} T$ are always invariant subspaces under a given T.
Note: it is also written "U is T-invariant" for some subspace U and transformation T.

Highly relevant to the concepts below: the COMP0147 notes on *cosets* and *quotient groups*.

Assume $U \subseteq V$ a subspace.

Coset For any $v \in V$, the coset $v + U$ (or $[v]$ in *equivalence class* notation) is the set $\{v + u \mid u \in U\}$.

Quotient Space A quotient (vector) space V/U is a new vector space over the set of all cosets of U.
I.e., V/U's carrier set is $\{\{v + u \mid u \in U\} \mid v \in V\}$.
Vector operations are defined as $[x] + [y] = [x + y]$ and $c[x] = [cx]$.
Observe that $\dim(V/U) = \dim V - \dim U$.

Note: there are no conditions on U; one can form a quotient vector space from *any* subspace. (Due to addition being abelian and multiplication being external.) This is contrary to general groups or rings, which require specific properties (normal subgroups or ideals) on the substructure.

Complementary Subspace A subspace $W \subseteq V$ is complementary to U iff. $V = U \oplus W$.
Construction from a Quotient Space:
Pick a basis $\{[v_1], ..., [v_k]\}$ for V/U.
Pick a $\tilde{v}_i \in [v_i]$ as a *representative* for every coset in the basis.
$W = \operatorname{span}\{\tilde{v}_1, ..., \tilde{v}_k\}$ is a subspace complementary to U.

Complements to a subspace are generally **non-unique**, and a complementary subspace can be viewed geometrically as a *section* of the corresponding quotient space. However, the addition of an inner product (to define orthogonality) allows the construction of a unique, **canonical** complement $U^\perp$.
(See *Orthogonality* subsection below.)

Observe that $W \cong (V/U)$ for any complement W. Specifically, the W-restriction of the projection map $\pi|_W : W \to V/U$ is an isomorphism.

The construction above produces a linear map $s : V/U \to V$ with $\pi \circ s = \mathrm{id}_{V/U}$, called a *section* (or *splitting*) of the projection $\pi : V \to V/U$. Its image is the complement W. Given a choice of s, the map $(u, [v]) \mapsto u + s([v])$ provides an *explicit* isomorphism $U \oplus (V/U) \cong V$. Because this isomorphism relies on the arbitrary choice of representatives to build s, there is no *canonical* isomorphism between V and $U \oplus (V/U)$.

4.2 Block Matrices

For a T-invariant U, one can construct, in a deliberate basis, a *block triangular* matrix.
Let W be a subspace complementary to U.

Pick a basis $B_U = (u_1, ..., u_k)$ for U and a basis $B_W = (w_1, ..., w_{n-k})$ for W.
$B = (u_1, ..., u_k, w_1, ..., w_{n-k})$ is then a basis for V. (Since $V = U \oplus W$.)

Now, we construct a matrix M for T in B.
Recall that a matrix can be constructed column-wise for $b \in B$ as $T(b)$'s coordinates (in B):

$$M = \begin{bmatrix} \uparrow & & \uparrow & \uparrow & & \uparrow \\ T(u_1) & \cdots & T(u_k) & T(w_1) & \cdots & T(w_{n-k}) \\ \downarrow & & \downarrow & \downarrow & & \downarrow \end{bmatrix} = \left[\begin{array}{c|c} A & B \\ \hline 0 & C \end{array}\right]$$

...where A is a $k \times k$ block; 0 is a $(n-k) \times k$ block of zeroes.

Observe:
- A is the B_U-coordinates of $T(U)$
- 0 is the B_W-coordinates of $T(U)$, which are zero since U is invariant
- B is the B_U-coordinates of $T(W)$
- C is the B_W-coordinates of $T(W)$

Observe that B will be zeroes if W is *also* invariant, yielding a *block diagonal* matrix.

Usefully, $\det(M) = \det(A) \times \det(C)$ and $\chi_M = \chi_A \times \chi_C$.

4.3 Eigenvalues and Eigenvectors

Eigenvector A nonzero vector $v \in V$ is an eigenvector of some T iff. there exists a scalar $\lambda \in \mathbb{F}$ where $T(v) = \lambda v$.

Eigenvalue The scalar λ above is the eigenvalue of T *corresponding* to the eigenvector v.

Other equations characterizing eigenvalues and eigenvectors wrt. matrix representations include

$$(M - \lambda I)v = \mathbf{0} \quad \text{or} \quad v \in \ker(M - \lambda I)$$

(See section below on the computational utility of these.)

An eigenvector's span is an invariant subspace where the associated endomorphism acts like a scaling: An eigenvector spans a 1-dimensional invariant subspace, and a linear transformation on 1-dimensional invariant subspaces can *only* be a scaling.

Eigenspace For a given eigenvalue λ, we have its eigenspace $E_\lambda = \ker(M - \lambda I) \subseteq V$.
Equivalently, $E_\lambda = \operatorname{span}\{v_1, ..., v_k\}$ where $\{v_1, ..., v_k\}$ is a basis of eigenvectors for λ (and k is the geometric multiplicity).

Observe that any $v \in E_\lambda \setminus \{\mathbf{0}\}$ is an eigenvector corresponding to λ.
Observe that $\{\text{eigenspaces of } T\} \subsetneq \{\text{invariant subspaces of } T\}$. (Unequal in nontrivial cases.)

Geometric Multiplicity The geometric multiplicity of an eigenvalue λ of M, $\gamma_M(\lambda)$, is $\dim E_\lambda$.
This is the "physical" dimension of the associated eigenspace.

Algebraic Multiplicity The algebraic multiplicity of an eigenvalue λ of M, $\alpha_M(\lambda)$, is its multiplicity as a root of χ_M (M's characteristic polynomial; see below).

Spectrum The spectrum of some T is the multiset of all its eigenvalues (with algebraic multiplicity).

Linear Independence of Eigenvectors For *distinct* eigenvalues $\lambda_1, ..., \lambda_k$, their corresponding eigenvectors $v_1, ..., v_k$ are linearly independent.
Alternatively: Take a basis for each eigenspace; the union of all these bases, across distinct eigenvalues, is linearly independent.
This allows the construction of bases out of eigenvectors if we have enough eigenvalues.

4.3.1 Computing Eigenvalues

Assume $v \in V \neq \mathbf{0}$ and $\lambda \in \mathbb{F}$.

Derivation from $Mv = \lambda v$ to the *characteristic polynomial*:

$$\begin{aligned} Mv &= \lambda v \\ Mv - \lambda v &= \mathbf{0} \\ Mv - \lambda(Iv) &= \mathbf{0} \\ (M - \lambda I)v &= \mathbf{0} \end{aligned}$$

Observe that a given λ is a solution
$\iff \ker(M - \lambda I)$ is non-trivial
$\iff M - \lambda I$ is not injective $\Leftrightarrow$ surjective $\Leftrightarrow$ bijective
$\iff M - \lambda I$ is not *invertible*
$\iff \det(M - \lambda I) = 0$.
I.e., any λ that satisfies any statement above is an eigenvalue, and the set of all such solutions are the eigenvalues of the transformation represented by M.

Characteristic Polynomial The characteristic polynomial χ_M for matrix M is

$$\chi_M(\lambda) = \det(M - \lambda I)$$

The roots of χ_M (expanding det with the Leibniz formula) are exactly the eigenvalues of M.

$$\begin{aligned} \chi_M(\lambda) = \det(\lambda I - M) &= \det\left(\begin{bmatrix} \lambda & 0 & \dots & 0 \\ 0 & \lambda & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda \end{bmatrix} - M\right) \\ &= \lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_2\lambda^2 + a_1\lambda + a_0 \end{aligned}$$

Aside²: there's a very subtle (and mostly insignificant) difference between $\det(M - \lambda I)$ and $\det(\lambda I - M)$. I refuse to elaborate further.

Observe that $T : V \to V$ cannot have more than $n = \dim V$ eigenvalues (or eigenspaces).
Observe that χ_M has degree n.
0 is a root of χ_M iff. $\det(M) = 0$

4.3.2 Computing Eigenvectors

Obviously, we just compute $\ker(M - \lambda I)$ for a given λ to obtain its eigenspace.

4.4 Polynomials and Linear Maps

The characteristic polynomial computes eigenvalues.
The roots and their multiplicity provide information on possible reductions (see below).

Complex vs Real fields
Per the fundamental theorem of algebra, not all roots of the characteristic polynomial may be real.
For $\mathbb{F} = \mathbb{R}$, some $n \times n$ matrix M has $\leq n$ eigenvalues.
For $\mathbb{F} = \mathbb{C}$, M has exactly n eigenvalues (counted with algebraic multiplicity).

Recall that polynomials in $\mathbb{R}$ factor into linear *and* irreducible quadratic factors, while polynomials in $\mathbb{C}$ factor fully into linear factors.

If $\mathbb{F} = \mathbb{R}$, χ_M factors into distinct irreducible quadratic terms P_i and linear terms.

$$\chi_M(x) = a \times P_1^{r_1}(x) \times \dots \times P_k^{r_k}(x) \times (x - \lambda_1)^{s_1} \times \dots \times \left(x - \lambda_j\right)^{s_j}$$

If $\mathbb{F} = \mathbb{C}$:

$$\chi_M(x) = a \times (x - \lambda_1)^{r_1} \times \dots \times (x - \lambda_m)^{r_m}$$

Annihilating Polynomial A polynomial P over $\mathbb{F}$ is an annihilating polynomial for M if $P(M) = 0$.

Minimal Polynomial The *monic* polynomial μ_M over $\mathbb{F}$ of least degree that annihilates M; $\mu_M(M) = 0$

Cayley-Hamilton Theorem Square matrices are annihilated by their own characteristic polynomials; $\forall M \in \mathcal{M}_{i,i} : \chi_M(M) = 0$
Effectively, this means $\{I, M, M^2, ..., M^n\}$ is linearly dependent, and χ_M itself demonstrates the exact dependence (via the coefficients). E.g., one can:
- compute matrix inverses cheaply (by rearranging $\chi_M(M)$)
- represent high powers M^k where $k \geq n$ as some linear combination of the lower powers

4.5 Matrix Reduction

Recall that a given matrix is just *one* representation of a linear transformation and is implicitly associated with a chosen basis.
Different bases thus correspond to different matrices that still represent the same transformation.

- We want diagonal matrices as they're computationally advantageous.
- While some T's matrix M in the standard basis may not be diagonal, it may have a diagonal representation under a different basis.
- Such a basis is precisely an *eigenbasis* – a basis consisting only of eigenvectors.

Eigenbasis For a transformation T, an *eigenbasis* is a basis of V consisting only of eigenvectors of T.

Diagonalization For a matrix M and its transformation T, M is *diagonalizable* iff. an eigenbasis exists for T.
Constructing P with the eigenbasis as columns, we obtain a change-of-basis matrix s.t. $D = P^{-1}MP$ is diagonal.

Further properties:
- the entries of D are the eigenvalues of M (with multiplicity)
- M is diagonalizable iff. μ_M splits into *distinct* linear factors.
- M is diagonalizable iff. $\alpha_M(\lambda) = \gamma_M(\lambda)$ for all eigenvalues λ.
- Powers of M are easy to compute: $M^k = PD^kP^{-1}$ (where D^k is just diagonal entries raised to k).
- Similarly, for any analytic function f (e.g., e^M): $f(M) = Pf(D)P^{-1}$.

Kernel Decomposition Theorem If the minimal polynomial of T is split into pairwise coprime factors $p(x) = p_1(x)^{r_1} \dots p_k(x)^{r_k}$, then V decomposes into a direct sum of the factors' kernels:

$$V = \ker(p_1(T)^{r_1}) \oplus \dots \oplus \ker(p_k(T)^{r_k})$$

Following the complete factorization of the characteristic polynomial, we have...
For $\mathbb{F} = \mathbb{R}$:

$$V = \ker(P_1^{r_1}(M)) \oplus \dots \oplus \ker(P_n^{r_k}(M)) \oplus \ker((M - \lambda_1 I)^{s_1}) \oplus \dots \oplus \ker\left(\left(M - \lambda_j I\right)^{s_j}\right)$$

For $\mathbb{F} = \mathbb{C}$:

$$V = \ker((M - \lambda_1 I)^{r_1}) \oplus \dots \oplus \ker((M - \lambda_m I)^{r_m})$$

Note that every term and its power r or s directly correspond to the factorization of χ_M.

Generalized Eigenspace For an eigenvalue λ with algebraic multiplicity $m = \alpha_M(\lambda)$, the generalized eigenspace is

$$K_\lambda = \ker((M - \lambda I)^m)$$

- Its elements are *generalized eigenvectors*.
- $\dim K_\lambda = \alpha_M(\lambda)$ (always, unlike eigenspaces where $\dim E_\lambda \leq \alpha_M(\lambda)$).
- While the direct sum of eigenspaces $\bigoplus E_\lambda$ may not span V, the direct sum of generalized eigenspaces always does (if χ_M splits, e.g., over $\mathbb{C}$):

$$V = \bigoplus_i K_{\lambda_i}$$

Triangularizable A matrix M is similar to an upper-triangular matrix iff. its characteristic polynomial χ_M splits fully into linear factors over $\mathbb{F}$.
- Since $\mathbb{C}$ is algebraically closed, all matrices over $\mathbb{F} = \mathbb{C}$ are upper-triangularizable.
- A real matrix is triangularizable over $\mathbb{R}$ iff. all its eigenvalues are real. If it has irreducible quadratic factors, it cannot be triangularized (but can be block-triangularized with 2×2 blocks).

Jordan Canonical Form Every matrix whose characteristic polynomial splits is similar to a block-diagonal matrix of *Jordan blocks*:

$$J = \begin{bmatrix} J_{d_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{d_2}(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{d_k}(\lambda_k) \end{bmatrix} \quad \text{where} \quad J_d(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{bmatrix}$$

5 Differential Calculus

5.1 Homogeneous Linear DEs

Homogeneous Linear Differential Equation (HLDE) A linear differential equation of the form

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + ... + a_1 y' + a_0 y = 0$$

Principle of superposition The linear combination of any two solutions to a HLDE is also a solution. This is due to the differential operator being linear.

The general solution to an n-th order HLDE is a linear combination of n linearly independent solutions.

Let $D = \frac{\mathrm{d}}{\mathrm{d}x}$ be the linear differential operator. The HLDE can be written as:

$$P(D)y = 0$$

where $P(z) = a_n z^n + ... + a_1 z + a_0$ is the characteristic polynomial.
Clearly, solving the HLDE is equivalent to solving for $\ker(P(D))$.

Spectral Mapping Theorem For any T and polynomial P, if v is an eigenvector of T with eigenvalue λ, then v is also an eigenvector of $P(T)$ with eigenvalue $P(\lambda)$:

$$Tv = \lambda v \quad \Longrightarrow \quad P(T)v = P(\lambda)v$$

Applying this to $T = D$, whose eigenfunctions are $y = e^{\lambda x}$ with eigenvalue λ (since $D(e^{\lambda x}) = \lambda e^{\lambda x}$), we get:

$$P(D)e^{\lambda x} = P(\lambda)e^{\lambda x}$$

Thus, if some λ satisfies $P(\lambda) = 0$, then $P(D)e^{\lambda x} = 0$, meaning the eigenfunction $e^{\lambda x} \in \ker(P(D))$ is a solution.

The general solution is built from the roots of the characteristic polynomial $\chi(\lambda) = P(\lambda) = 0$:

1. **Distinct Real Roots**: For each real root λ_i, there is a solution $e^{\lambda_i x}$.
2. **Repeated Real Roots**: For a real root λ with multiplicity m, the kernel contains the m linearly independent solutions:

$$e^{\lambda x}, x e^{\lambda x}, ..., x^{m-1} e^{\lambda x}$$

3. **Complex Conjugate Roots**: Since the coefficients a_i are real, complex roots appear in conjugate pairs $\lambda = a \pm ib$.
   - These yield complex solutions $e^{(a \pm ib)x} = e^{ax} e^{\pm ibx} = e^{ax}(\cos(bx) \pm i \sin(bx))$.
   - Taking the real and imaginary parts (which are linear combinations) yields two real, linearly independent solutions:

$$e^{ax}\cos(bx) \quad \text{and} \quad e^{ax}\sin(bx)$$

   - If repeated with multiplicity m, multiply these by powers x^k for $k < m$.

For complex conjugate solutions $y_1 = u + iv$ and $y_2 = u - iv$, their real and imaginary parts are linear combinations of the conjugate pair themselves:

$$u = \frac{y_1 + y_2}{2} \quad \text{and} \quad v = \frac{y_1 - y_2}{2i}$$

By superposition, these linear combinations are also solutions, and they are guaranteed to be real-valued and linearly independent.

5.2 Multivariate Functions

Fubini's Theorem For a function $f : \mathbb{R}^2 \to \mathbb{R}$ that is continuous over a rectangle $X \times Y$ (X and Y are intervals), the double integral can be computed as an iterated integral:

$$\iint_{X \times Y} f(x,y)\,\mathrm{d}(x,y) = \int_X \left(\int_Y f(x,y)\,\mathrm{d}y \right) \mathrm{d}x = \int_Y \left(\int_X f(x,y)\,\mathrm{d}x \right) \mathrm{d}y$$

This double-to-iterated conversion is valid for any shape of area as long as f is continuous over it.

Gradient A representation of the first derivative of a multivariate function.
For a function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient of f at some point $x = (x_1, ..., x_n) \in R^n$ is the vector of partial derivatives:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ ... \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

The gradient points in the direction of steepest ascent, and its magnitude is the rate of increase in that direction.

Hessian A representation of the second derivative of a multivariate function.
For a function $f : \mathbb{R}^n \to \mathbb{R}$, the Hessian of f at some point $x = (x_1, ..., x_n) \in R^n$ is the matrix of second partial derivatives:

$$(\mathcal{H}_f)_{i,j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

$$\mathcal{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} = \mathbf{J}(\nabla f)^\top$$

The Hessian is used to analyze the local curvature of f and classify stationary points.

5.3 Stationary Points

If the second partial derivatives of f are continuous near a point x, then $\frac{\partial^2 f}{\partial x \partial y}(x) = \frac{\partial^2 f}{\partial y \partial x}(x)$.
Consequently, the Hessian is symmetric for any f with continuous second partial derivatives.

Classification of Stationary point x:

- $\det(\mathcal{H}_f(x)) > 0$
  - If $\frac{\partial^2 f}{\partial x_1^2}(x) > 0$, then x is a local minimum.
  - If $\frac{\partial^2 f}{\partial x_1^2}(x) < 0$, then x is a local maximum.
- $\det(\mathcal{H}_f(x)) < 0$
  x is a saddle point.
- $\det(\mathcal{H}_f(x)) = 0$
  inconclusive

6 Euclidean Spaces

Continue assuming V is a finite-dimensional vector space over $\mathbb{F} = \mathbb{R}$ *or* $\mathbb{F} = \mathbb{C}$;
$u, v, w \in V$ and $a, b \in \mathbb{F}$.

6.1 Inner Products

The *canonical* inner product is the dot product.

Inner Product Additional structure on a V as a binary operation $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ satisfying:
1. Conjugate symmetry: $\langle v, w \rangle = \overline{\langle w, v \rangle}$
2. Linearity in the first argument: $\langle av + bw, u \rangle = a\langle v, u \rangle + b\langle w, u \rangle$
3. Positive-definiteness: $u \neq 0 \Longrightarrow \langle u, u \rangle > 0$

Observe that for $\mathbb{F} = \mathbb{R}$, $x = \overline{x}$, so 1. is reduced to simple symmetry, and 2. extends to bilinearity.
Also, for $\mathbb{F} = \mathbb{C}$, 1. implies that $\langle u, u \rangle \in \mathbb{R}$.

Implied properties:
- $\langle 0, u \rangle = \langle u, 0 \rangle = 0$
- $\langle u, u \rangle = 0 \iff u = 0$ (A way to prove zero-ness of a vector)
- $\langle u, av + bw \rangle = \overline{a}\langle u, v \rangle + \overline{b}\langle u, w \rangle$; i.e., conjugate-linearity in the second argument (*sesquilinear*).

The inner product is a *symmetric positive-definite bilinear form* for $\mathbb{F} = \mathbb{R}$, and a *positive-definite sesquilinear form* for $\mathbb{F} = \mathbb{C}$.

Inner Product Space A vector space equipped with an inner product.

Euclidean Space An inner product space over $\mathbb{R}$.

Matrix Representation of Inner Product A matrix $M \in \mathcal{M}_{n,n}$ represents the inner product

$$\langle u, v \rangle = u^\top M v$$

for $u, v \in \mathbb{R}^n$ iff. M is symmetric and positive-definite; i.e.,
- $M^\top = M$
- $v \neq 0 \Longrightarrow v^\top M v > 0$

For $\mathbb{F} = \mathbb{C}$, replace "transposition" $x^\top$ with "conjugate transposition" $x^\dagger = \overline{x}^\top$ and these conditions remain true.

I.e., (in finite dimension), every inner product can be represented as a matrix, and every matrix satisfying said conditions represents an inner product.

$\langle \cdot, \cdot \rangle_M$ denotes the inner product represented by M.

6.1.1 Infinite Dimension

Let's for example consider functions $f, g : \mathbb{R} \to \mathbb{R}$ on the interval $[a, b]$.

The *canonical* inner product is

$$\langle f, g \rangle = \int_a^b f(x)g(x)\,\mathrm{d}x$$

Symmetry comes from the commutativity of multiplication;
linearity is inherited from that of integration;
positive-definiteness is inherited from $\mathbb{R}$ as a group.

6.2 Norms

Norm A function $\|\cdot\| : V \to \mathbb{R}$ satisfying:
1. Subadditivity (triangle inequality): $\|v + w\| \leq \|v\| + \|w\|$
2. Absolute homogeneity: $\|av\| = |a|\ \|v\|$
3. Positive-definiteness: $u \neq 0 \Longrightarrow \|u\| > 0$

Induced Norm Every inner product naturally *induces* a norm via

$$\|u\| = \sqrt{\langle u, u \rangle}$$

Note that the proof of the triangle inequality relies on the Cauchy-Schwarz inequality (below).

The other direction, i.e., whether a norm induces an inner product, requires the *parallelogram law* $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$.

Euclidean Norm The norm induced by the canonical inner product in Euclidean spaces. E.g.,

$$\left\| \begin{bmatrix} x \\ y \end{bmatrix} \right\| = \sqrt{\left\langle \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle} = \sqrt{\begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}} = \sqrt{x^2 + y^2}$$

Cauchy-Schwarz Inequality In any inner product space V,
$|\langle v, w \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle$
or, assuming the induced norm,
$|\langle v, w \rangle| \leq \|v\|\|w\|$.

Note that the Cauchy-Schwarz inequality is a consequence of the axioms of inner products; it is not an axiom itself, but it is a non-trivial result that helps prove, e.g., $\sqrt{\langle u, u \rangle}$ is indeed a norm.

6.3 Metrics

Metric (function) (Aka. distance) A function $d : A \times A \to \mathbb{R}$ satisfying:
1. Symmetry: $d(x, y) = d(y, x)$
2. Positive-definiteness: $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$
3. Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$

for all $x, y, z \in A$.

Metric Space A structure (A, d) where $d : A \times A \to \mathbb{R}$ is a metric.

Induced Metric Inner product spaces induce the metric $d(u, v) = \|u - v\|$.

Every inner product space is canonically a metric space using its induced norm and thus induced metric.

6.4 Orthogonality

Orthogonality For an inner product space, two vectors are orthogonal $v \perp w$ iff. $\langle v, w \rangle = 0$.

Generalized Pythagorean Theorem For an inner product space with the induced norm $\|\cdot\|$,

$$v \perp w \quad \Longrightarrow \quad \|v + w\|^2 = \|v\|^2 + \|w\|^2$$

Orthonormal Basis A basis where vectors are pairwise orthogonal and of unit norm.

The coordinates of a vector in an orthonormal basis are easily computed via the inner product; for a basis $(b_1, \ldots, b_n)$ and a vector v, $v = \langle v, b_1 \rangle b_1 + \ldots + \langle v, b_n \rangle b_n$.

Every orthonormal family of vectors is linearly independent, and every finite-dimensional inner product space has an orthonormal basis.

Orthogonal Complementary Subspace A given inner product subspace $W \subseteq V$ has a unique orthogonal complement (aka., the canonical complement)

$$W^\perp = \{v \in V \mid \langle v, w \rangle = 0\ \forall w \in W\}$$

Clearly,
$V^\perp = \{0\}$ and $\{0\}^\perp = V$
$V = W + W^\perp$

For an orthonormal basis $(w_1, \ldots, w_k)$ of $W \subseteq V$, we can define the *orthogonal projection* s.t.

$$P_W(v) = \langle v, w_1 \rangle w_1 + \ldots + \langle v, w_k \rangle w_k$$

Clearly, $P_W(v) \in W$ and $v - P_W(v) \in W^\perp$.

Orthogonal Matrix A matrix $M \in \mathcal{M}_{n,n}$ is orthogonal iff. its columns form an orthonormal basis.
Alternatively, $M^\top M = I$ or $M^\top = M^{-1}$.

Real Spectral Theorem For a real symmetric matrix M,
- M has a full eigenbasis
- The eigenbasis can be chosen to be orthonormal

Consequently, M is orthogonally diagonalizable: there exists an orthogonal matrix P s.t.
$D = P^\top M P$ is diagonal.

For **real symmetric matrices**,
- eigenvectors are orthogonal (and eigenvalues are real)
- positive-definiteness $\iff$ (strictly) positive eigenvalues (**useful property**)

This eigenvalue condition for positive-definiteness allows the *easier* classification of matrices as inner products.

Recall that matrix representations of inner products are symmetric;
they can thus always be diagonalized.
Furthermore, positive eigenvalues allows normalization of an diagonal form into the identity matrix, so **every inner product can be represented as the canonical inner product in some basis**.

Advantages of an orthonormal basis:
1. Coordinates are easily computed via the inner product.
2. Inner products become the canonical dot product.
3. Orthogonal projections have a closed form (inner product).
4. Change of basis matrices between orthonormal bases are orthogonal.
5. A bunch more…

7 Numerical Methods

7.1 Newton's Method

finds roots...

Prerequisites, algorithm + convergence

7.2 Euler's Method

solves ODEs...

Algorithm + error bounding

7.3 Gradient Descent

finds minima...

algorithm + convergence

7.4 Numerical Integration

integrates...

7.4.1 Riemann Sums

rectangles, trapezoids... (brief metion of simpson's rule)

7.4.2 Monte Carlo

8 Probabilities

8.1 Basics

Random Variable A function $X : \Omega \to T$ where <insert measure theory baggage>

> To properly define a *random variable*, we need some background in *measure theory* (or some setup with commutative unital algebras)... but that is completely out of scope... *gives up*

Sample Space Ω, the set of possible outcomes.
Target Space T, the image of X. Often, $T = \mathbb{R}^n$.

When T is continuous, X is a *continuous random variable.*

Probability Density Function (PDF) Any function $f : T \to \mathbb{R}$ where

1. $\forall x \in T : f(x) \geq 0$
2. $\int_T f(x)\,\mathrm{d}x = 1$

A given random variable X can have multiple PDFs, but with enough measure theory knowledge you know there is a canonical one :).

If a random variable X has PDF f, then for any $a, b \in T$ with $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x$$

Note that it does not matter whether the inequalities are strict or not; they're all equivalent for continuous random variables since $\Pr(X = a) = 0$ for any a.

Cumulative Distribution Function (CDF) For a random variable X, its CDF is $F : T \to \mathbb{R}$ where

$$F(x) = P(X \leq x)$$

If X has a PDF f, then also

$$F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$$

Inverse CDF For a "distribution" $\mathcal{X}$, we notate its inverse CDF as $\mathcal{X}[c]$. I.e.,
$\mathcal{X}[c] = x \iff P(X \leq x) = c$.

Expected Value (Mean) For a random variable X with PDF f, the expected value is

$$\mu = E[X] = \int_T x f(x)\,\mathrm{d}x$$

If X is discrete, then $E[X] = \sum_T x P(X = x)$.

> Note that the expected value (and thus variance) can be defined for any random variable, even without a PDF, but this requires measure theory; for any random variable X on $(\Omega, \mathcal{F}, P)$,
>
> $$E[X] = \int_\Omega X(\omega)\,\mathrm{d}P(\omega)$$
>
> In the following section on statistics, this justifies the computation of means and variances on arbitrary statistical samples (which are assumed to be of some unspecified random variable).

Covariance For random variables X and Y with means μ_X and μ_Y and *joint* PDF $f_{X,Y}$,

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y] \\ &= \int_T \int_T (x - \mu_X)(y - \mu_Y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y\end{aligned}$$

> The "product" XY is a new random variable of the pointwise product of X and Y, i.e., $(XY)(\omega) = X(\omega)Y(\omega)$ for all $\omega \in \Omega$.
>
> But of course this doesn't directly yield useful computational formulas... so you should learn measure theory.

Variance For a random variable X with PDF f and mean μ,

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{Cov}(X,X) \\ &= E[(X - E[X])^2] \\ &= E[X^2] - E[X]^2 \\ &= \int_T x^2 f(x)\,\mathrm{d}x - \mu^2 = \int_T (x - \mu)^2 f(x)\,\mathrm{d}x\end{aligned}$$

Covariance measures the dependence between X and Y;
variance measures the spread of variable around its mean.

> **Law of the Unconscious Statistician** For a random variable X with PDF f and any function $g : T \to U$ (U may be different target space),
>
> $$E[g(X)] = \int_T g(x) f(x)\,\mathrm{d}x$$
>
> This follows from the definition of the expected value but is not immediately obvious.
> This explains why $E[X^2] = \int_T x^2 f(x)\,\mathrm{d}x$ above.

Standard Deviation $\sigma(X) = \sqrt{\mathrm{Var}(X)}$

If X and Y are independent, then $\mathrm{Cov}(X,Y) = 0$.

Independence X and Y are independent iff. $P(X = x \cap Y = y) = P(X = x)P(Y = y)$ (for all x, y).

For independent X and Y, the joint PDF factorizes: $f_{X,Y}(x,y) = f_X(x) f_Y(y)$.
Also, $E[XY] = E[X]E[Y]$ and thus $\mathrm{Cov}(X,Y) = 0$.

8.2 Classic Distributions

8.2.1 Exponential

$$\begin{aligned}X &\sim \mathrm{Exp}(\lambda) && \{\lambda \in \mathbb{R}^+\} \\ f(x) &= \lambda e^{-\lambda x} && \{x \geq 0\} \\ T &= [0, \infty) \\ E[X] &= \frac{1}{\lambda} \\ \mathrm{Var}(X) &= \frac{1}{\lambda^2}\end{aligned}$$

8.2.2 Uniform

$$\begin{aligned}X &\sim U(a,b) && \{a < b \in \mathbb{R}\} \\ f(x) &= \frac{1}{b-a} && \{a \leq x \leq b\} \\ T &= [a,b] \\ E[X] &= \frac{a+b}{2} \\ \mathrm{Var}(X) &= \frac{(b-a)^2}{12}\end{aligned}$$

8.2.3 Normal

$$\begin{aligned}X &\sim N(\mu, \sigma^2) && \{\mu \in \mathbb{R}, \sigma \in \mathbb{R}^+\} \\ f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \\ T &= \mathbb{R} \\ E[X] &= \mu \\ \mathrm{Var}(X) &= \sigma^2\end{aligned}$$

Aka. Gaussian distribution.
No closed-form CDF.

8.2.4 Student's t-Distribution

$$\begin{aligned}X &\sim t(\nu) && \{\nu \in \mathbb{R}^+\} \\ f(x) &= \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}} \\ T &= \mathbb{R} \\ E[X] &= \begin{cases} 0 & \text{for } \nu > 1 \\ \text{undefined} & \text{otherwise} \end{cases} \\ \mathrm{Var}(X) &= \begin{cases} \frac{\nu}{\nu-2} & \text{for } \nu > 2 \\ \infty & \text{for } 1 < \nu \leq 2 \\ \text{undefined} & \text{otherwise} \end{cases}\end{aligned}$$

Its sole parameter ν is called the *degrees of freedom* and is usually an integer.

Similar to the normal distribution but with heavier tails.
As $\nu \to \infty$, $t(\nu) \to N(0,1)$.

Definition via χ^2 and normal distributions:

$$\frac{Z}{\sqrt{V/\nu}} \sim t(\nu)$$

where $Z \sim N(0,1)$ and $V \sim \chi^2(\nu)$ (and are independent).

8.2.5 Chi-Squared

$$\begin{aligned}X &\sim \chi^2(k) && \{k \in \mathbb{Z}^+\} \\ f(x) &= \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}}\,\Gamma(k/2)} && \{x > 0\} \\ T &= [0, \infty) \\ E[X] &= k \\ \mathrm{Var}(X) &= 2k\end{aligned}$$

Its sole parameter k is called the *degrees of freedom.*

$$\sum_{i=1}^{k} Z_i^2 \sim \chi^2(k)$$

where $Z_i \overset{\text{iid}}{\sim} N(0,1)$.

$\chi^2(k) = \mathrm{Gamma}\left(\alpha = \frac{k}{2}, \theta = 2\right)$

8.3 Multivariate

...joint probability distributions...

$$P(X = a, Y = b) = P(X = a \cap Y = b)$$

Joint distro of two continuous variables:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x,y)\,\mathrm{d}y\,\mathrm{d}x$$

for the *joint* PDF $f_{X,Y}$ of X and Y.

Marginal Distributions Of a multivariate distribution, a single-variate distributions at some fixed values of all other variables. E.g. for the joint PDF $f_{X,Y}$,

$$\begin{aligned}f_X(x) &= \int_T f_{X,Y}(x,y)\,\mathrm{d}y \\ f_Y(y) &= \int_T f_{X,Y}(x,y)\,\mathrm{d}x\end{aligned}$$

(Like a 2-d cross-section of the 3-d joint graph...)

Conditional Probabilities

$$\begin{aligned}P(X \leq x \mid Y = y) &= \frac{P(X \leq x \cap Y = y)}{P(Y = y)} \\ &= \frac{\int_{-\infty}^{x} f_{X,Y}(t,y)\,\mathrm{d}t}{f_Y(y)}\end{aligned}$$

8.4 Useful Stuff

Law of Large Numbers For a sequence of *independent and identically distributed* (i.i.d.) random variables, the average converges to the individual expected value as the number of variables goes to infinity.
Formally, if $X_1, X_2, ..., X_n$ are i.i.d. with expected value μ, then

$$P\left(\lim_{n\to\infty} \overline{X} = \mu\right) = 1$$

where $\overline{X} = \frac{1}{n}(X_1 + ... + X_n)$.

Central Limit Theorem (CLT) For i.i.d.s $X_1, X_2, ..., X_n$ with mean μ and *finite* variance σ^2,

$$\lim_{n\to\infty} \overline{X} \sim N(\mu, \sigma^2/n)$$

where $\overline{X} = \frac{1}{n}(X_1 + ... + X_n)$.

We consider the approximation valid for $n \geq 30$.

9 Statistics

Statistics can be seen as the physical manifestation of associated theoretical concepts in probability.

- Interpreting probabilities as statistical quantities

  Interesting: see *frequentist* vs *Bayesian* interpretations of probability.

9.1 The Normal Distribution

The normal distribution frequently appears in various statistical processes.
Different processes may produce different normal distributions, but they can be normalized to the *standard* normal distribution $N(\mu = 0, \sigma^2 = 1)$.

Normalization For some $X \sim N(\mu, \sigma^2)$, the normalized variable $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$.

z	$P(\lvert Z\rvert \leq z)$
1	68.27%
1.96	95.00%
2	95.45%
3	99.73%

9.2 Parameter Estimation

To model a process with a given distribution, we must estimate its defining parameters from sample data. E.g., for a normal distribution, we estimate the mean μ and variance σ^2; for an exponential distribution, we estimate the rate λ.

If a realized sample of n observations is drawn from the distribution of one random variable X, we can model the sample prior to realization as a random sample: a sequence of i.i.d.s $X_1, ..., X_n$ sharing the same distribution as X.
Per the CLT, the distribution of the sample mean $\overline{X}$ is approximated by a normal distribution (for sufficient n).
It is therefore useful to estimate the μ and σ^2 of an arbitrarily-distributed variable X, as these two parameters directly dictate the behavior of the sample mean even when X itself is not normal.

Sample (Random Sample) A sample $\boldsymbol{X}$ of size n is a sequence of random variables $\boldsymbol{X} = (X_1, ..., X_n)$.
A *realized* sample is an observation of $\boldsymbol{X}$, i.e. a sequence of real values $\boldsymbol{x} = (x_1, ..., x_n)$ where x_i is the observed realization of X_i.

Simple Random Sample (SRS) A sample where X_i are i.i.d.

Clearly, the CLT applies to the sample mean of SRSs.

Statistic Any function T that maps a random sample $\boldsymbol{X}$ to a *statistic* (a new random variable).
When evaluated on a realized sample, $T(\boldsymbol{x})$ yields a real value.

E.g., the sample mean estimator $\overline{X} = \frac{1}{n}\sum_i X_i$ is a random variable, while the realized sample mean $\overline{x} = \frac{1}{n}\sum_i x_i$ is a fixed number.

Estimator An estimator $T(\boldsymbol{X})$ is a statistic used to estimate a parameter θ of the underlying distribution. The realized value $\hat{\theta} = T(\boldsymbol{x})$ is an **estimate**.

Bias An estimator $T(\boldsymbol{X})$ for θ is unbiased iff. $E[T(\boldsymbol{X})] = \theta$.

E.g. the sample mean estimator $\overline{X}$ is an unbiased estimator of the true mean μ since $E[\overline{X}] = \mu$.

Sampling Distribution The sampling distribution of an estimator $T(\boldsymbol{X})$ is its probability distribution as a random variable.
Equivalently, it is the distribution of realized estimates $T(\boldsymbol{x})$ across all possible realizations of the random sample $\boldsymbol{X}$.

Knowledge of the sampling distribution of an estimator allows us to e.g. compute *confidence intervals* for the estimate, and thus is a crucial part of statistics.

Note that statistics and their sampling distributions are usually tied to the sample size n.

Parent Normality Assumption: Note that the exact mathematical derivations for the sampling distributions of the estimated mean and variance (specifically, their exact connection to Student's t-distribution and the χ^2-distribution) below strictly depend on the **assumption that the underlying parent distribution is normal** (i.e., $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$). If the underlying distribution is non-normal, these exact finite-sample results do not hold.
Nevertheless, for the sample mean, the CLT provides an asymptotic approximation as $n \to \infty$ independent of the underlying distribution. No such general non-parametric distribution holds for the sample variance in finite samples, motivating alternative non-parametric approaches like bootstrapping below.

9.2.1 Mean

Mean Estimator The canonical sample mean estimator is the arithmetic mean:

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

It is **unbiased**:

$$E[\overline{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}(n\,E[X]) = \frac{1}{n}(n\mu) = \mu$$

Also, $\operatorname{Var}(\overline{X}) = E\left[\left(\overline{X} - \mu\right)^2\right] = \frac{\sigma^2}{n}$.
Recall that in general, $E[\overline{X}] = \mu$ and $\operatorname{Var}(\overline{X}) = \frac{\sigma^2}{n}$.
Under the parent normality assumption $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, the exact sampling distribution of the sample mean is

$$\overline{X} \sim N(\mu, \sigma^2/n)$$

which can be standardized:

$$Y = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\frac{\overline{X} - \mu}{\sigma} \sim N(0, 1)$$

...and rearranged to yield the realized $(1 - \alpha)$ confidence interval for μ:

$$\left[\overline{x} + z\left[\frac{\alpha}{2}\right]\frac{\sigma}{\sqrt{n}}, \overline{x} + z\left[1 - \frac{\alpha}{2}\right]\frac{\sigma}{\sqrt{n}}\right] \quad \text{or} \quad \left[\overline{x} - \left|z\left[\frac{\alpha}{2}\right]\right|\frac{\sigma}{\sqrt{n}}, \overline{x} + \left|z\left[\frac{\alpha}{2}\right]\right|\frac{\sigma}{\sqrt{n}}\right]$$

($z[c]$ is the inverse CDF of the standard normal distribution.)

Studentized Mean Often, the true variance σ^2 required in the above interval is unknown. If σ is substituted with the unbiased sample standard deviation S, the resulting statistic instead follows Student's t-distribution with $n - 1$ degrees of freedom:

$$Z = \sqrt{n}\frac{\overline{X} - \mu}{S} \sim t(n - 1)$$

Because we are using an estimator S instead of a constant σ, Z is the quotient of a normal distribution (the mean estimator) and the square root of a chi-squared distribution (the variance estimator, see below), which results in a t-distribution. The t-distribution exhibits heavier tails than the normal distribution to account for this uncertainty in the variance.

Using this exact sampling distribution, the $(1 - \alpha)$ realized confidence interval for μ is now

$$\left[\overline{x} - \left|t_{n-1}\left[\frac{\alpha}{2}\right]\right|\frac{s}{\sqrt{n}}, \overline{x} + \left|t_{n-1}\left[\frac{\alpha}{2}\right]\right|\frac{s}{\sqrt{n}}\right]$$

where $\overline{x}$ and s are the mean and standard deviation of the realized sample.

Clearly, $t_{n-1}\left[\frac{\alpha}{2}\right] \geq z\left[\frac{\alpha}{2}\right]$, so the realized confidence intervals are wider when the variance is also estimated.

9.2.2 Variance

Variance Estimator The estimator for sample variance following the discrete formula is

$$\tilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$$

However, $\tilde{S}^2$ is a **biased** estimator of the population variance σ^2 (and thus called *uncorrected*):

$$E\left[\tilde{S}^2\right] = \text{...complex derivation...} = \frac{n-1}{n}\sigma^2$$

The bias arises because the sample mean $\overline{X}$ is used in place of the unknown true population mean μ. The sample observations X_i are closer to their own sample mean $\overline{X}$ than to the true mean μ, causing $\tilde{S}^2$ to systematically underestimate the true variance.

Bessel's Correction $\tilde{S}^2$'s bias is corrected by scaling by $\frac{n}{n-1}$, yielding the **unbiased sample variance estimator**:

$$S^2 = \frac{n}{n-1}\tilde{S}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$$

Now,

$$E[S^2] = \frac{n}{n-1}E\left[\tilde{S}^2\right] = \frac{n}{n-1}\left(\frac{n-1}{n}\sigma^2\right) = \sigma^2$$

We denote the corresponding realized estimates (calculated from a realized sample $\boldsymbol{x}$) as $\tilde{s}^2$ and s^2.

Under the parent normality assumption ($X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$), the sampling distribution of S^2 is related to the chi-squared χ^2 distribution.
Specifically, the standardized statistic K follows a χ^2 distribution with $n - 1$ degrees of freedom:

$$K = \frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2 \sim \chi^2(n-1)$$

(Note that χ^2 is asymmetric.)
For a confidence level of $1 - \alpha$, we have:

$$P\left(\chi^2_{n-1}\left[\frac{\alpha}{2}\right] \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{n-1}\left[1 - \frac{\alpha}{2}\right]\right) = 1 - \alpha$$

$$P\left(\frac{(n-1)S^2}{\chi^2_{n-1}\left[1 - \frac{\alpha}{2}\right]} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{n-1}\left[\frac{\alpha}{2}\right]}\right) = 1 - \alpha$$

For a realized sample with unbiased variance estimate s^2, the confidence interval for σ^2 is

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1}\left[1 - \frac{\alpha}{2}\right]}, \frac{(n-1)s^2}{\chi^2_{n-1}\left[\frac{\alpha}{2}\right]}\right]$$

To obtain the realized confidence interval for the standard deviation σ, we simply take the square root of the bounds.

9.3 Hypothesis Testing

Hypothesis testing provides a structured statistical framework to make decisions about population parameters using sample data.

9.3.1 Core Concepts

Null Hypothesis (H_0) The baseline or default assumption representing "no change," "no effect," or the status quo. Generally written as $H_0 : \mu = \mu_0$.

Alternative Hypothesis (H_1) The claim we seek to establish evidence for, contradicting H_0. It can take three forms:
- **Two-tailed:** $H_1 : \mu \neq \mu_0$ (testing for any deviation from μ_0)
- **Left-tailed:** $H_1 : \mu < \mu_0$ (testing if parameter is strictly smaller)
- **Right-tailed:** $H_1 : \mu > \mu_0$ (testing if parameter is strictly larger)

Test Statistic A standardized sample statistic whose distribution is known under the null hypothesis. For a normal population with unknown variance, we calculate the observed t-statistic:

$$t_0 = \frac{\overline{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \sqrt{n}\frac{\overline{x} - \mu_0}{s}$$

Under H_0, this statistic is a realization of the random variable $T \sim t(n - 1)$.

p-value The probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the observed value t_0. The calculation of "extreme" depends directly on H_1:
- **Two-tailed:** $p = P(|T| \geq |t_0|) = 2P(T \geq |t_0|)$
- **Left-tailed:** $p = P(T \leq t_0)$
- **Right-tailed:** $p = P(T \geq t_0)$

Significance Level (d or α) The maximum probability threshold (typically 5%) of rejecting H_0 when it is actually true (Type I error).
- If $p < d$: we **reject $\boldsymbol{H_0}$** in favor of H_1 (the observed effect is statistically significant).
- If $p \geq d$: we **fail to reject $\boldsymbol{H_0}$** (we cannot conclude anything significant; we do **not** "accept" H_0, but rather find insufficient evidence to discard it).

9.3.2 Critical Value Approach

An equivalent decision method relies on comparing the observed test statistic t_0 with critical values t_{crit} from the t-table for a given significance level α:
- **Two-tailed:** Reject H_0 if $|t_0| \geq t_{1-\frac{\alpha}{2}, n-1}$
- **Left-tailed:** Reject H_0 if $t_0 \leq -t_{1-\alpha, n-1}$
- **Right-tailed:** Reject H_0 if $t_0 \geq t_{1-\alpha, n-1}$

9.3.3 Concrete Example: Knee Surgery Recovery Times

A researcher investigates whether going to physical therapy 3 times a week (instead of the usual 2) shortens knee surgery recovery times. The default average recovery time is $\mu_0 = 8$ weeks.

1. **Formulate Hypotheses:**

$$H_0 : \mu = 8 \text{ weeks}$$
$$H_1 : \mu < 8 \text{ weeks} \qquad \{\text{Left-tailed test}\}$$

2. **Collect Sample Data:** A sample of $n = 12$ patients undergoing therapy 3 times a week yields a sample mean of $\overline{x} = 7.2$ weeks with a sample standard deviation of $s = 1.5$ weeks.
3. **Compute the Test Statistic:**

$$t_0 = \sqrt{12}\frac{7.2 - 8}{1.5} \approx -1.848$$

4. **Calculate the p-value:** With $n - 1 = 11$ degrees of freedom, the left-tailed p-value is:

$$p = P(T \leq -1.848) = P(T \geq 1.848) = 1 - P(T \leq 1.848)$$

Consulting the t-distribution table for $d.f. = 11$, we find the cumulative probability $P(T \leq 1.848) \approx 0.9503$:

$$p \approx 1 - 0.9503 = 0.0497$$

5. **Make Decision (at 5% significance level):** Since $p = 0.0497 < 0.05$, we reject the null hypothesis H_0. **Conclusion:** There is statistically significant evidence that physical therapy 3 times a week shortens recovery time compared to the standard 2 times.

9.4 Bootstrapping

Traditional statistical inference relies on parametric assumptions (e.g., population normality) or large-sample asymptotics (the CLT) to analytically derive the sampling distribution of an estimator. However, in many practical scenarios, these models fail:
- The underlying distribution may be highly skewed, multi-modal, or simply non-normal (e.g., alcohol consumption or wealth distributions).
- The sample size n may be too small for asymptotic theorems (like the CLT) to provide a reliable approximation.
- Sourcing additional physical samples may be impossible or prohibitively expensive (e.g., a one-off historical survey).

Bootstrapping provides a computational method to estimate the sampling distribution of any statistic directly from a single observed sample.
The only assumption is that the observed sample is *representative* of the underlying distribution.

9.4.1 Bootstrap Algorithm

Given:
- An unknown population distribution with a true parameter of interest θ.
- An observed sample $\boldsymbol{x} = (x_1, ..., x_n)$ of size n.
- An statistic $T(\boldsymbol{X})$ estimating θ, yielding an initial estimate $\hat{\theta} = T(\boldsymbol{x})$.

Repeat B times:
1. Draw a uniform random sample $\boldsymbol{x}^* = (x_1^*, ..., x_n^*)$ of size n from $\boldsymbol{x}$ uniformly **with replacement**.
2. Compute the statistic $\hat{\theta}^* = T(\boldsymbol{x}^*)$.

This yields a sequence of B bootstrap estimates: $\hat{\theta}_1^*, \hat{\theta}_2^*, ..., \hat{\theta}_B^*$.
As $B \to \infty$, the empirical distribution of the bootstrap estimates converges to the true sampling distribution of $\hat{\theta}$, allowing statistical inference (e.g. confidence intervals) without parametric assumptions.

9.4.2 Constructing Bootstrap Confidence Intervals

Under the **Percentile Bootstrap Method**, we can construct a $(1 - \alpha)$ confidence interval for θ directly from the empirical distribution of our bootstrap estimates:
1. Sort the B bootstrap estimates $\hat{\theta}_i^*$ in increasing order.
2. Find the lower percentile index at $\frac{\alpha}{2}$ and the upper percentile index at $1 - \frac{\alpha}{2}$.
3. The resulting confidence interval is

$$\left[\hat{\theta}^*_{[B\,\frac{\alpha}{2}]}, \hat{\theta}^*_{[B\,(1-\frac{\alpha}{2})]}\right]$$